

Task-1

```
1  #include<iostream>
2  using namespace std;
3  class Calculator{
4      public:
5      int multiply(int a,int b ){
6          return a*b;
7      }
8      int multiply(int a,int b,int c){
9          return a*b*c;
10     }
11     }
12     double multiply(double a,double b){
13         return a*b;
14     }
15 };
16 int main(){
17     Calculator obj;
18     cout << obj.multiply(2,3) << endl;
19     cout << obj.multiply(2,3,4) << endl;
20     cout << obj.multiply(22.3,33.2) << endl;
21 }
```

```
6
24
740.36
```

Task-2

```

1  #include<iostream>
2  using namespace std;
3  class shape{
4  public:
5      void draw(){
6          cout<<"Draw the Shape"<<endl;
7      }
8  };
9  class circle:public shape{
10 public:
11     void draw(){
12         cout<<"Draw the circle"<<endl;
13     }
14 };
15
16 class rectangle:public shape{
17 public:
18     void draw(){
19         cout<<"Draw the rectangle"<<endl;
20     }
21 };
22 int main(){
23     circle c;
24     c.draw();
25     rectangle r;
26     r.draw();
27 }

```

Draw the circle
Draw the rectangle

```

1  #include<iostream>
2  using namespace std;
3  class Shape{
4      public:
5          int area(int side){
6              return side*side;
7          }
8          int area(int length,int width){
9              return length*width;
10         }
11         double area(double radius){
12             return 3.14*radius*radius;
13         }
14     };
15
16 int main(){
17     Shape square;
18     Shape rectangle;
19     Shape circle;
20     cout<<"Area of Square: "<<square.area(25)<<endl;
21     cout<<"Area of Rectangle: "<<rectangle.area(12,2)<<endl;
22     cout<<"Area of circle: "<<circle.area(3.50)<<endl;
23
24 }

```

```

Area of Square: 625
Area of Rectangle: 24
Area of circle: 38.465

```

Task-4

```
1  #include<iostream>
2  using namespace std;
3  class Employee{
4      public:
5      void CalculateSalary(){
6          cout<<"Base Class"<<endl;
7      }
8  };
9  class Manager :public Employee{
10     public:
11     void CalculateSalary(){
12         cout<<"Manager Salary: 22500"<<endl;
13     }
14 };
15 class Developer: public Employee{
16     public:
17     void CalculateSalary(){
18         cout<<"Developer Salary: 25000"<<endl;
19     }
20 };
21 class Intern:public Employee{
22     public:
23     void CalculateSalary(){
24         cout<<"Intern Salary: 14000"<<endl;
25     }
26 };
27 int main(){
28     Manager m;
29     m.CalculateSalary();
30     Developer d;
31     d.CalculateSalary();
32     Intern i;
33     i.CalculateSalary();
34
35 }
```

Manager Salary: 22500
Developer Salary: 25000
Intern Salary: 14000

Task-5

```
1  #include<iostream>
2  using namespace std;
3  class Complex{
4      private:
5          double real,img;
6      public:
7          Complex(double r,double i):real(r),img(i){}
8
9      Complex operator + (Complex b){
10         Complex c(0,0);
11         c.real = real + b.real;
12         c.img = img + b.img;
13         return c;
14     }
15     Complex operator - (Complex b){
16         Complex c(0,0);
17         c.real = real - b.real;
18         c.img = img - b.img;
19         return c;
20     }
21     Complex operator * (Complex b){
22         Complex c(0,0);
23         c.real = real * b.real;
24         c.img = img * b.img;
25         return c;
26     }
27     void GetValues(){
28         cout<<real<<"+"<<img<<"i"<<endl;
29     }
30 };
31
```

```
int main(){
    Complex a(2,4);
    Complex b(3,4);
    Complex c = a+b;
    c.GetValues();
    Complex d = b-a;
    d.GetValues();
    Complex e = a*b;
    e.GetValues();
}
```

```
5+8i
1+0i
6+16i
```

Task-6

```
1  #include<iostream>
2  using namespace std;
3  class Vehicle{
4      public:
5      virtual void startEngine(){
6          cout<<"Vehicle Engine Started"<<endl;
7      }
8  };
9  class Car:public Vehicle{
10     public:
11     void startEngine(){
12         cout<<"Car Engine Started"<<endl;
13     }
14 };
15 class Bike:public Vehicle{
16     public:
17     void startEngine(){
18         cout<<"Bike Engine Started"<<endl;
19     }
20 };
21 class Truck:public Vehicle{
22     public:
23     void startEngine(){
24         cout<<"Truck Engine Started"<<endl;
25     }
26 };
```

```
int main(){
    Vehicle *ptr;
    Bike b;
    Car c;
    Truck t;
    ptr = &b;
    ptr->startEngine();
    ptr = &c;
    ptr->startEngine();
    ptr = &t;
    ptr->startEngine();
}
```

Bike Engine Started
Car Engine Started
Truck Engine Started